

Metric 1 / 3 Action probe top-1 accuracy

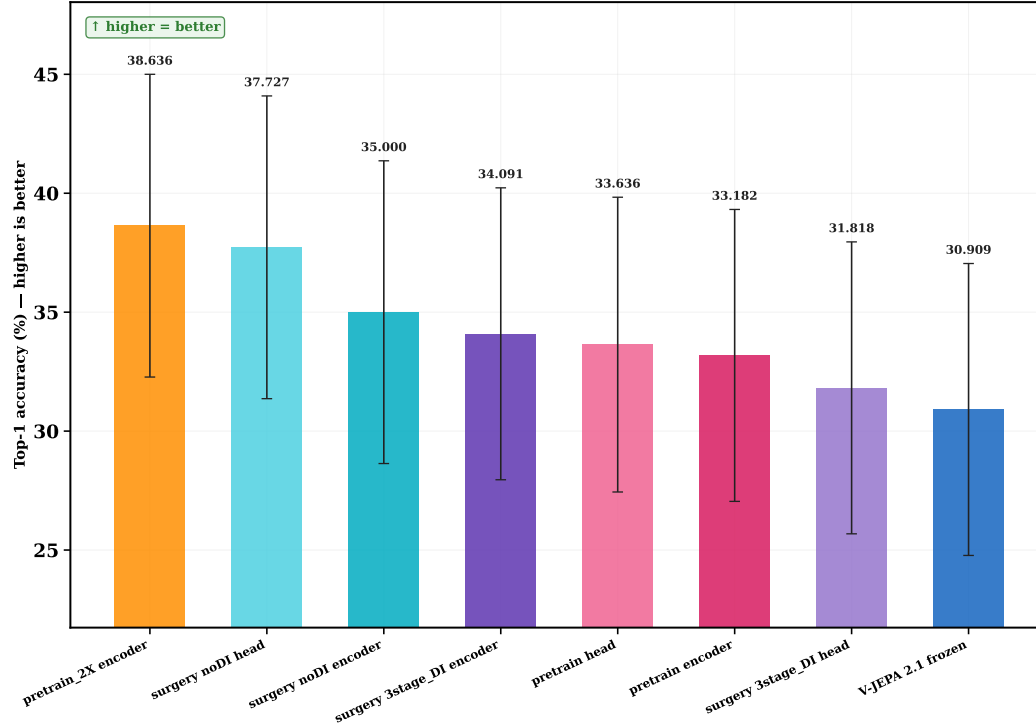
Linear separability of motion-action classes on encoder features (higher is better)

v1 | n = 220 | test-clip leakage

v2 | n = 1825 | leakage fixed

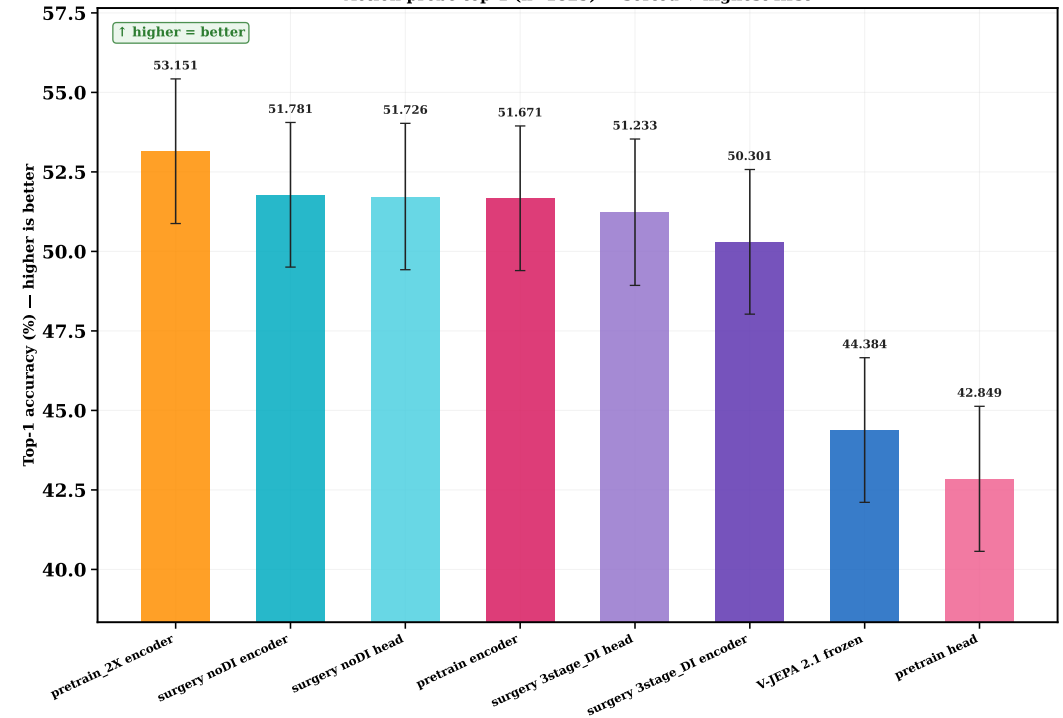
Action probe top-1 (n=220) — sorted ↓ highest first · 8 encoders · 95 % BCa CI

Action probe top-1 (n=220) — sorted ↓ highest first



Action probe top-1 (n=1825) — sorted ↓ highest first · 8 encoders · 95 % BCa CI

Action probe top-1 (n=1825) — sorted ↓ highest first



TAKEAWAY

1. v2 removes test-clip leakage and grows the eval set 220 -> 1825 clips; 95% CI half-width: 6.1 -> 2.3 pp (~2.7x, the 1/sqrt(N) law).
2. Encoder adaptation lifts top-1 to 50-53% - CI-separated +6 to +9 pp above frozen (44.4%) in v2.
3. Surgery vs continual-pretrain: top-six CIs OVERLAP in BOTH versions - no separation, no leakage-driven reversal.

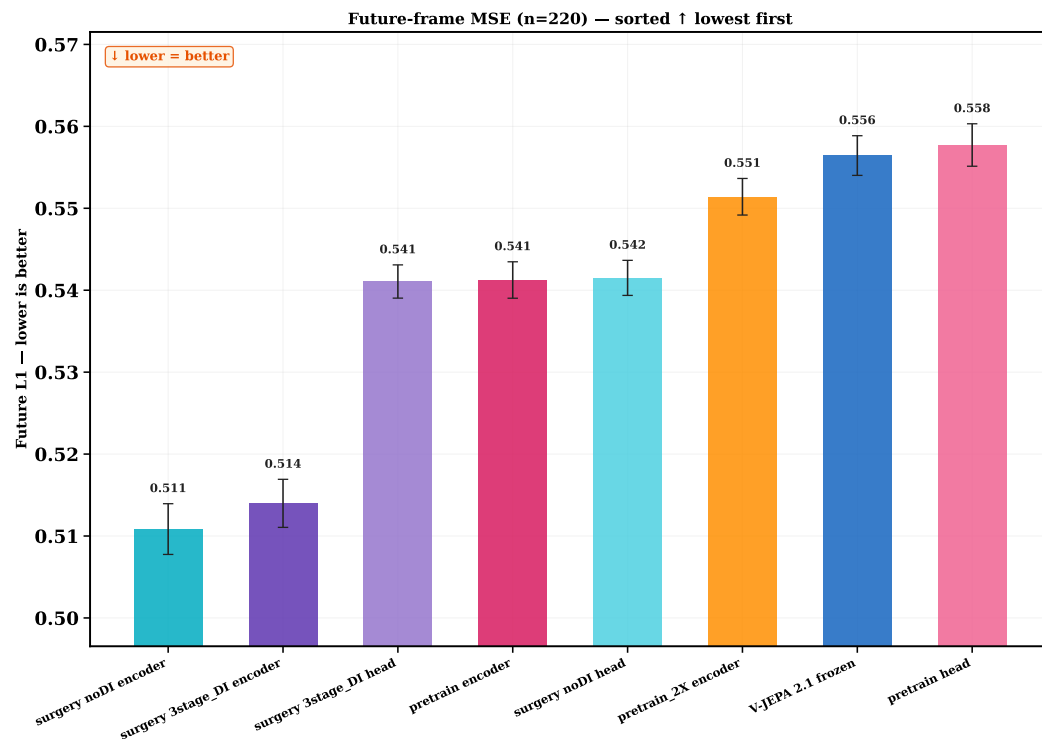
Metric 2 / 3 Future-frame prediction error (L1)

Predictor rollout error on held-out future frames (lower is better)

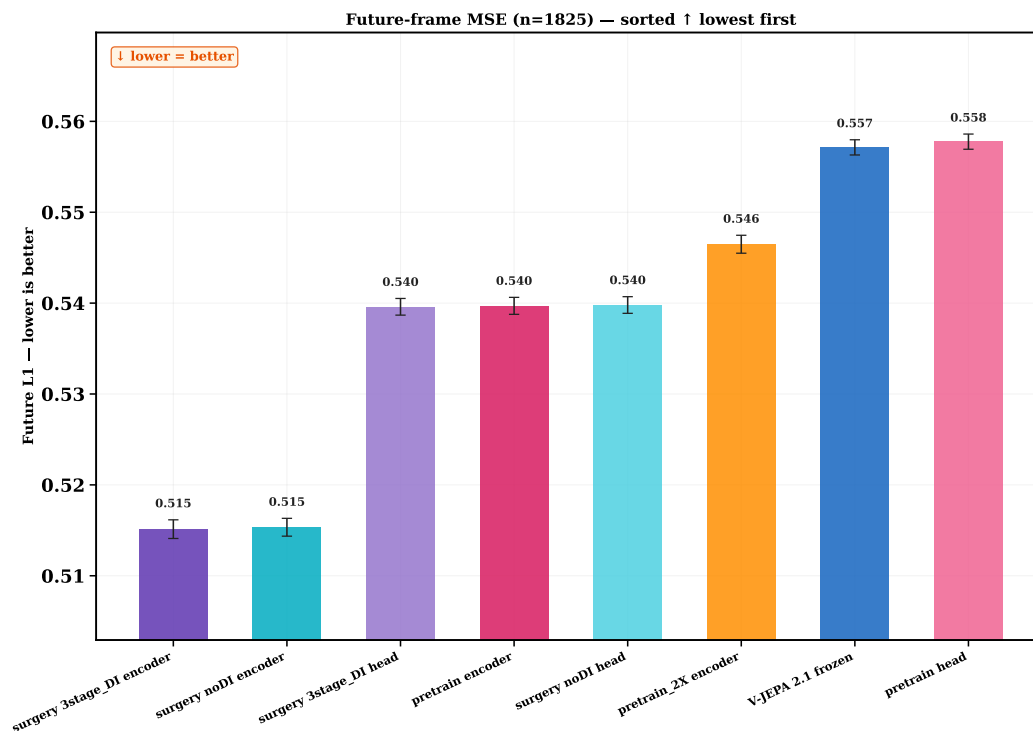
v1 | n = 220 | test-clip leakage

v2 | n = 1825 | leakage fixed

Future-frame MSE (n=220) — sorted ↑ lowest first · 8 encoders · 95 % BCa CI



Future-frame MSE (n=1825) — sorted ↑ lowest first · 8 encoders · 95 % BCa CI



TAKEAWAY

1. Same leakage fix + 8x eval clips; 95% CI half-width: 0.0024 -> 0.0008 (~2.9x).
2. Cleanest win: surgery-on-encoder gives the LOWEST error (0.515), CI-separated below all pretrain (≥ 0.540) and frozen (0.557) - both versions.
3. The gain is encoder-specific: surgery-on-head ~ continual-pretrain (~0.540).

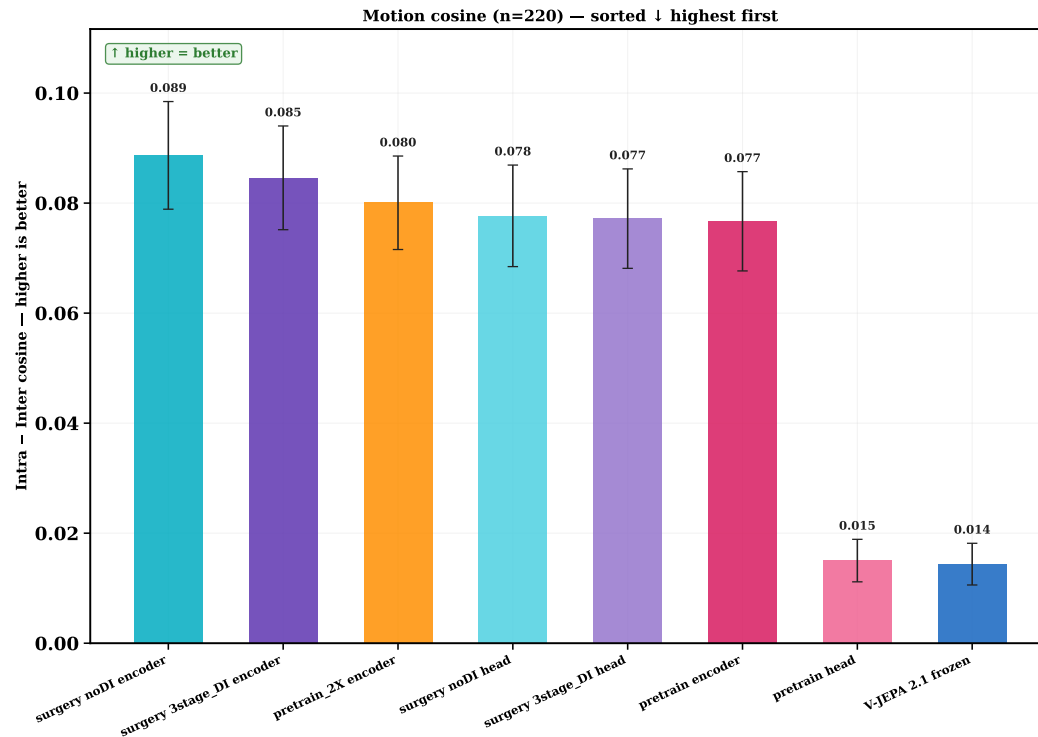
Metric 3 / 3 Motion cosine: intra-inter class separation

Same-class minus different-class cosine of motion features (higher is better)

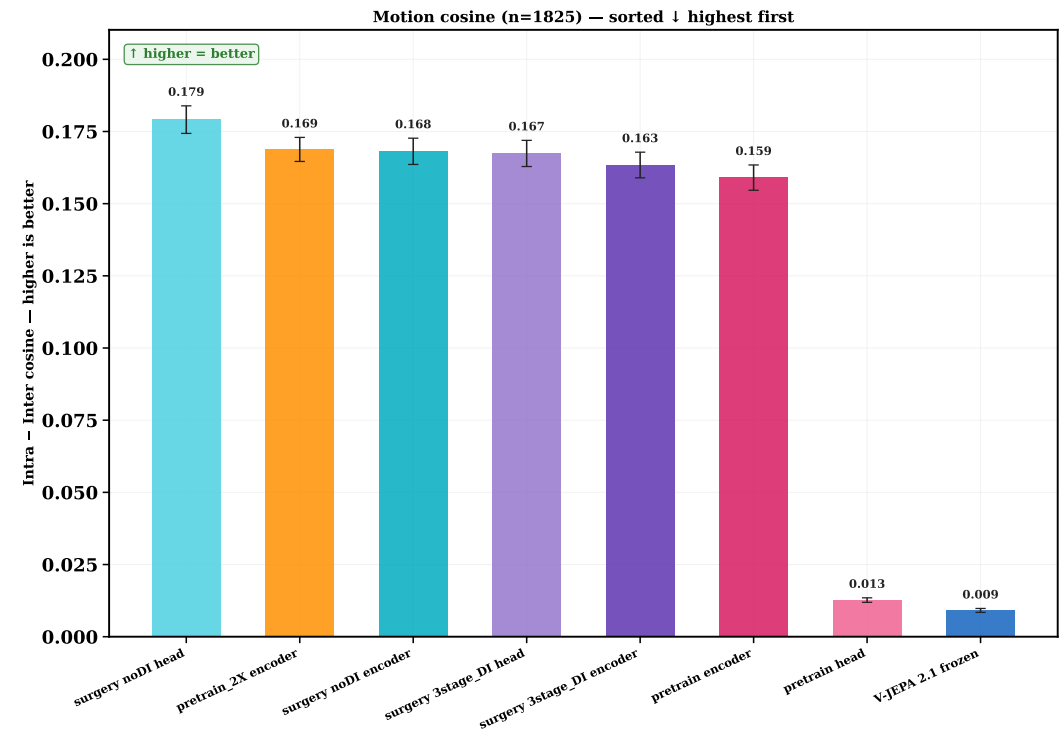
v1 | n = 220 | test-clip leakage

v2 | n = 1825 | leakage fixed

Motion cosine (n=220) — sorted ↓ highest first · 8 encoders · 95 % BCa CI



Motion cosine (n=1825) — sorted ↓ highest first · 8 encoders · 95 % BCa CI



TAKEAWAY

1. Same fix + 8x clips; with the clean larger set the class separation roughly DOUBLES (~0.08 -> 0.17).
2. Encoder adaptation creates motion structure: 6 adapted variants at 0.16-0.18; frozen & head-only pretrain stay near 0 (~0.01).
3. Best surgery (noDI-head 0.179) > best pretrain (2X 0.169): CI-separated in v2, overlapping CIs in v1.